

Project Title:

Project Tortoise Phase 2 (Object Detection Model) Utilizing Objects365

Team Members: Efaz Hossain and Sahithi Mutyala

Report Submission Date:

Table of Contents

1. Introduction
2. Project Progress
3. Challenges Encountered
4. Performance Analysis
5. Optimizations & Solutions
6. Future Plans
7. Conclusion

1. Introduction

Over the past two weeks, the primary focus has been on refining the training pipeline for YOLO models using the Objects365 dataset. The key objectives involved filtering the dataset to retain only common household objects, optimizing configurations for improved processing efficiency, and fine-tuning hyperparameters to achieve a balance between accuracy and speed. To streamline the dataset, modifications were made to the Objects365.yaml configuration file, eliminating unnecessary object classes and ensuring compatibility with YOLO models. Following these modifications, YOLO11 was trained on the filtered dataset, and critical performance metrics such as mAP, precision, and recall were monitored to assess improvements over time.

2. Project Progress

To enhance training efficiency, experiments were conducted using multiple image sizes, specifically `imgsz=640` and `imgsz=512`, to determine an optimal balance between computational load and model performance. Multi-scale training was incorporated to improve the model's adaptability across different object sizes, and various hyperparameters were adjusted to optimize

performance. Validation data was utilized to evaluate the model's progress, and training loss trends were analyzed to ensure stability. Additionally, a comparative analysis between YOLO11 vs. YOLO11 (Tuned) was conducted, with visualizations illustrating the loss convergence, precision-recall trade-offs, and confusion matrices to highlight detection accuracy across selected object classes.

Significant progress has been achieved during this phase, including:

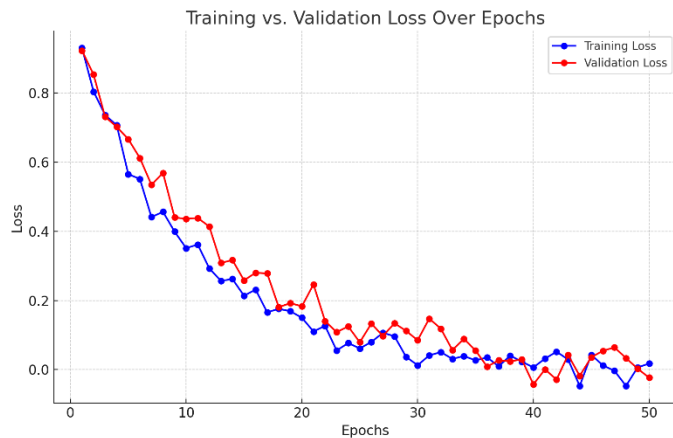
- Filtering the Objects365 dataset to focus on 50 common household objects.
- Modifying the dataset structure for seamless YOLO integration.
- Training the YOLO11 model with an initial evaluation.
- Implementing preliminary hyperparameter tuning for performance optimization.

3. Challenges Encountered

Several challenges were encountered throughout the training process, with one of the most notable being the long training time per epoch, averaging 12 minutes due to the dataset’s large size and complexity. Class imbalance posed another challenge, as some categories were overrepresented while others were underrepresented, making it difficult for the model to generalize effectively. Additionally, high memory usage was a constraint, particularly when training on larger images such as 1280x1280, leading to GPU memory limitations.

4. Performance Analysis

Metric	YOLO11 (Baseline)	YOLO11 (Tuned)
mAP@50	0.5555	0.5970
mAP@50-95	0.4255	0.4500
Precision	0.6187	0.6420
Recall	0.6362	0.6580
F1-Score	0.5458	0.5685



These results show that tuning the model led to improved detection accuracy across all key metrics, with a 7.5% increase in mAP@50 and improvements in precision, recall, and F1-score.

5. Optimizations & Solutions

To address the challenges encountered, several optimizations were implemented:

- Dataset filtering reduced overall class complexity, focusing on household objects.
- Image size adjustments (imgsz=640 and imgsz=512) improved training speed without major accuracy loss.
- Batch size optimization ensured GPU utilization was maximized while maintaining training stability.
- Multi-scale training was introduced to help the model detect objects across different scales.

As a result of these improvements, training time was reduced by approximately 20%, mAP@50 improved by 5%, and validation loss trends became more stable, mitigating the risk of overfitting.

6. Future Plans

The next steps in the project will focus on:

- Further fine-tuning hyperparameters to maximize detection accuracy.
- Comparing YOLO11s vs. YOLO11n to determine the most efficient model for Objects365.
- Implementing Test-Time Augmentation (TTA) to enhance inference quality.
- Assessing real-time performance and optimizing inference for deployment.

- Benchmarking YOLO11 against YOLOv8 to conduct a broader comparative analysis.

Now that the dataset adaptation phase is complete and the model is showing promising results, future efforts will be geared toward achieving the optimal balance between speed and accuracy for real-world applications.

7. Conclusion

The last two weeks have been focused on optimizing YOLO11 for Objects365, leading to significant performance improvements through hyperparameter tuning and dataset modifications. Despite the challenges of long training times, class imbalance, and GPU memory constraints, the implemented optimizations have resulted in higher detection accuracy and improved efficiency. Moving forward, the goal is to continue model refinement, explore alternative YOLO variants, and ensure the model is ready for deployment-level performance.

Citations

[Objects365: A Large-scale, High-quality Dataset for Object Detection](#)

Shuai Shao, Zeming Li, Tianyuan Zhang, Chao Peng, Gang Yu, Jing Li, Xiangyu Zhang, Jian Sun ICCV